

Suggested FYP topics for 2026-27

by

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Project code :	TFX1
Project title :	Burst photography for high dynamic range and low-light imaging
No. of students :	2
Description of project :	Many mobile cameras nowadays provide a night-time mode to capture images in lowlight. However, images captured in lowlight conditions are still very noisy and low-quality. In this project, we will explore how to burst photography to improve the images captured in lowlight conditions. Students are expected to build a burst photography and to evaluate images captured by different cameras. Students are expected to be familiar with Python and Numpy and better to take image and video processing and machine learning courses before.
Industrial collaboration :	No.

Project code :	TFX 2
Project title :	Creative editing using depth map estimated from a single image
No. of students :	2
Description of project :	Depth map is very important information for creative image editing, like generating synthetic background blur (Bokeh effect) or 3D motion photo. In this project, we will use machine learning algorithms to estimate depth of objects from images. Furthermore, utilizing the estimated depth map, we will further use them to generate creative image editing effects, including Bokeh and 3D motion photo. Students are expected to be familiar with Python and Numpy and better to take image and video processing and machine learning courses before.
Industrial collaboration :	No

Project code :	TFX 3
Project title :	Image editing and quality assessment via foundation models
No. of students :	1
Description of project :	Image editing and assessment are important topics in image processing. Traditional editing or assessment algorithms are designed for each individual task, like denoising or blurriness assessment. In this project, the students are expected to explore the utilization of large foundation models, including stable diffusion-based image generation (AIGC models) and multi-modality language models to solve multiple image editing and assessment tasks. Students are expected to be familiar with Python and Numpy and better to take image and video processing and machine learning courses before. Students should also have own account for any commercial MLLM tool, like ChatGPT 4.0.
Industrial collaboration :	No.